

- ✓ Program - The current population of a town is 10000. The population of the town is increasing at the rate of 10% per year. You have to write a program to find out the population at the end of each of the last 10 years.

```
current_population = 10000
current_year = 2010
year = (current_year - 10)
while current_year > year:
    print(f"In {current_year} population of town is {round(current_population,2)}")
    current_population = current_population/1.1
    current_year = current_year - 1
```

```
↩ In 2010 population of town is 10000
In 2009 population of town is 9090.91
In 2008 population of town is 8264.46
In 2007 population of town is 7513.15
In 2006 population of town is 6830.13
In 2005 population of town is 6209.21
In 2004 population of town is 5644.74
In 2003 population of town is 5131.58
In 2002 population of town is 4665.07
In 2001 population of town is 4240.98
```

Sequence sum

$1/1! + 2/2! + 3/3! + \dots$

✓ Nested Loops

```
# Examples -> unique pairs
for i in range(1,5):
    for j in range(1,5):
        print(i,j)
```

```
↩ 1 1
1 2
1 3
1 4
2 1
2 2
2 3
2 4
3 1
3 2
3 3
3 4
4 1
4 2
4 3
4 4
```

✓ Pattern 1

```
*
**
***
```

```
for i in range(1,4):
    for j in range(1,i+1):
        print("*",end=" ")
    print()
```

```
↩ *
  * *
 * * *
```

✓ Pattern 2

```
1
121
12321
1234321
```

```
for i in range(1,5):
    for j in range(1,i+1):
        print(j,end=" ")
    for k in range(i-1,0,-1):
        print(k,end=" ")
    print()
```

```
↔ 1
   1 2 1
   1 2 3 2 1
   1 2 3 4 3 2 1
```

✓ Loop Control Statement

- Break
- Continue
- Pass

```
for i in range(1,10):
    if i == 5:
        break
    print(i)
```

```
↔ 1
   2
   3
   4
```

```
lower = int(input('enter lower range'))
upper = int(input('enter upper range'))
```

```
for i in range(lower,upper+1):
    for j in range(2,i):
        if i%j == 0:
            break
    else:
        print(i)
```

```
↔ enter lower range10
   enter upper range100
   11
   13
   17
   19
   23
   29
   31
   37
   41
   43
   47
   53
   59
   61
   67
   71
   73
   79
   83
   89
   97
```

```
# Continue
for i in range(1,10):
    if i == 5:
        continue
    print(i)
```

```
↕ 1
   2
   3
   4
   6
   7
   8
   9
```

```
for i in range(1,10):
    pass
```

Strings are sequence of Characters

In Python specifically, strings are a sequence of Unicode Characters

- Creating Strings
- Accessing Strings
- Adding Chars to Strings
- Editing Strings
- Deleting Strings
- Operations on Strings
- String Functions

✓ Creating Stings

```
s = 'hello'
s = "hello"
# multiline strings
s = '''hello'''
s = """hello"""
s = str('hello')
print(s)
```

```
↕ hello
```

```
"it's raining outside"
```

```
↕ 'it's raining outside'
```

✓ Accessing Substrings from a String

```
# Positive Indexing
s = 'hello world'
print(s[41])
```

```
↕ -----
IndexError                                Traceback (most recent call last)
<ipython-input-61-633ba99ed6e5> in <module>
      1 # Positive Indexing
      2 s = 'hello world'
----> 3 print(s[41])

IndexError: string index out of range
```

```
# Negative Indexing
s = 'hello world'
print(s[-3])
```

```
↕ r
```

```
# Slicing
s = 'hello world'
print(s[6:0:-2])
```

```
↕ wol
```

```
print(s[::-1])
```

```
↵ dlrow olleh
```

```
s = 'hello world'
print(s[-1:-6:-1])
```

```
↵ dlrow
```

✓ Editing and Deleting in Strings

```
s = 'hello world'
s[0] = 'H'
```

Python strings are immutable

```
↵ -----
TypeError                                 Traceback (most recent call last)
<ipython-input-80-0c8a824e3b73> in <module>
      1 s = 'hello world'
----> 2 s[0] = 'H'

TypeError: 'str' object does not support item assignment
```

```
s = 'hello world'
del s
print(s)
```

```
↵ -----
NameError                                 Traceback (most recent call last)
<ipython-input-81-9ae37fbf1c6c> in <module>
      1 s = 'hello world'
      2 del s
----> 3 print(s)

NameError: name 's' is not defined
```

```
s = 'hello world'
del s[-1:-5:2]
print(s)
```

```
↵ -----
TypeError                                 Traceback (most recent call last)
<ipython-input-82-d0d823eafb6b> in <module>
      1 s = 'hello world'
----> 2 del s[-1:-5:2]
      3 print(s)

TypeError: 'str' object does not support item deletion
```

✓ Operations on Strings

- Arithmetic Operations
- Relational Operations
- Logical Operations
- Loops on Strings
- Membership Operations

```
print('delhi' + ' ' + 'mumbai')
```

```
↵ delhi mumbai
```

```
print('delhi'*5)
```

```
↵ delhidelhidelhidelhidelhi
```

```
print("*"*50)
```

```
➞ *****
```

```
'delhi' != 'delhi'
```

```
➞ False
```

```
'mumbai' > 'pune'  
# lexicographically
```

```
➞ False
```

```
'Pune' > 'pune'
```

```
➞ False
```

```
'hello' and 'world'
```

```
➞ 'world'
```

```
'hello' or 'world'
```

```
➞ 'hello'
```

```
'' and 'world'
```

```
➞ ''
```

```
'' or 'world'
```

```
➞ 'world'
```

```
'hello' or 'world'
```

```
➞ 'hello'
```

```
'hello' and 'world'
```

```
➞ 'world'
```

```
not 'hello'
```

```
➞ False
```

```
for i in 'hello':  
    print(i)
```

```
➞ h  
   e  
   l  
   l  
   o
```

```
for i in 'delhi':  
    print('pune')
```

```
➞ pune  
   pune  
   pune  
   pune  
   pune
```

```
'D' in 'delhi'
```

```
➞ False
```

✓ Common Functions

- len
- max
- min
- sorted

```
len('hello world')
```

```
↔ 11
```

```
max('hello world')
```

```
↔ 'w'
```

```
min('hello world')
```

```
↔ ' '
```

```
sorted('hello world',reverse=True)
```

```
↔ ['w', 'r', 'o', 'o', 'l', 'l', 'l', 'h', 'e', 'd', ' ']
```

✓ Capitalize/Title/Upper/Lower/Swapcase

```
s = 'hello world'
print(s.capitalize())
print(s)
```

```
↔ Hello world
   hello world
```

```
s.title()
```

```
↔ 'Hello World'
```

```
s.upper()
```

```
↔ 'HELLO WORLD'
```

```
'Hello WoLrd'.lower()
```

```
↔ 'hello woLrd'
```

```
'HeLlO WorLD'.swapcase()
```

```
↔ 'hElLo wOrld'
```

✓ Count/Find/Index

```
'my name is nitish'.count('i')
```

```
↔ 3
```

```
'my name is nitish'.find('x')
```

↔ -1

```
'my name is nitish'.index('x')
```

↔

```
-----  
ValueError                                Traceback (most recent call last)  
<ipython-input-121-12e2ad5b75e9> in <module>  
----> 1 'my name is nitish'.index('x')  
  
ValueError: substring not found
```

✓ endswith/startswith

```
'my name is nitish'.endswith('sho')
```

↔ False

```
'my name is nitish'.startswith('1my')
```

↔ False

✓ format

```
name = 'nitish'  
gender = 'male'
```

```
'Hi my name is {1} and I am a {0}'.format(gender,name)
```

↔ 'Hi my name is nitish and I am a male'

✓ isalnum/ isalpha/ isdigit/ isidentifier

```
'nitish1234%'.isalnum()
```

↔ False

```
'nitish'.isalpha()
```

↔ True

```
'123abc'.isdigit()
```

↔ False

```
'first-name'.isidentifier()
```

↔ False

✓ Split/Join

```
'hi my name is nitish'.split()
```

↔ ['hi', 'my', 'name', 'is', 'nitish']

```
" ".join(['hi', 'my', 'name', 'is', 'nitish'])
```

↔ 'hi my name is nitish'

✓ Replace

```
'hi my name is nitish'.replace('nitish', 'campusx')
```

```
→ 'hi my name is nitish'
```

✓ Strip

```
'nitish'.strip()
```

```
→ 'nitish'
```

✓ Example Programs

```
# Find the length of a given string without using the len() function
```

```
s = input('enter the string')
```

```
counter = 0
```

```
for i in s:  
    counter += 1
```

```
print('length of string is', counter)
```

```
→ enter the stringnitish  
length of string is 6
```

```
# Extract username from a given email.  
# Eg if the email is nitish24singh@gmail.com  
# then the username should be nitish24singh
```

```
s = input('enter the email')
```

```
pos = s.index('@')  
print(s[:pos])
```

```
→ enter the emailsupport@campusx.in  
support
```

```
# Count the frequency of a particular character in a provided string.  
# Eg 'hello how are you' is the string, the frequency of h in this string is 2.
```

```
s = input('enter the email')  
term = input('what would like to search for')
```

```
counter = 0  
for i in s:  
    if i == term:  
        counter += 1
```

```
print('frequency', counter)
```

```
→ enter the emailhi how are you  
what would like to search foro  
frequency 2
```

```
# Write a program which can remove a particular character from a string.
```

```
s = input('enter the string')  
term = input('what would like to remove')
```

```
result = ''
```

```
for i in s:  
    if i != term:  
        result = result + i
```

```
print(result)
```

```
→ enter the stringnitish
   what would like to removei
   ntsh
```

```
# Write a program that can check whether a given string is palindrome or not.
# abba
# malayalam
```

```
s = input('enter the string')
flag = True
for i in range(0,len(s)//2):
    if s[i] != s[len(s) - i -1]:
        flag = False
        print('Not a Palindrome')
        break
```

```
if flag:
    print('Palindrome')
```

```
→ enter the stringpython
   Not a Palindrome
```

```
# Write a program to count the number of words in a string without split()
```

```
s = input('enter the string')
L = []
temp = ''
for i in s:
```

```
    if i != ' ':
        temp = temp + i
    else:
        L.append(temp)
        temp = ''
```

```
L.append(temp)
print(L)
```

```
→ enter the stringhi how are you
   ['hi', 'how', 'are', 'you']
```

```
# Write a python program to convert a string to title case without using the title()
s = input('enter the string')
```

```
L = []
for i in s.split():
    L.append(i[0].upper() + i[1:].lower())

print(" ".join(L))
```

```
→ enter the stringhi my namE iS NitiSh
   Hi My Name Is Nitish
```

```
# Write a program that can convert an integer to string.
```

```
number = int(input('enter the number'))
```

```
digits = '0123456789'
result = ''
while number != 0:
    result = digits[number % 10] + result
    number = number//10
```

```
print(result)
print(type(result))
```

```
→ enter the number345
   345
```

<class 'str'>